

PROPOSITION

Let X_1, X_2 be conti. rand vars on a prob space.

Suppose we define $Y_1 = g_1(X_1, X_2)$ and $Y_2 = g_2(X_1, X_2)$, where $g_1, g_2: \mathbb{R}^2 \rightarrow \mathbb{R}$

If $\forall (y_1, y_2) \in \mathbb{R}^2, \exists! (x_1, x_2) \in \mathbb{R}^2$, s.t. $y_1 = g_1(x_1, x_2), y_2 = g_2(x_1, x_2)$ and $J(x_1, x_2) = \frac{\partial(g_1, g_2)}{\partial(x_1, x_2)} \neq 0$, then $f_{Y_1, Y_2}(y_1, y_2) = f_{X_1, X_2}(x_1(y_1, y_2), x_2(y_1, y_2)) |J(x_1, x_2)|^{-1}$

Proof

Consider the joint CDF for Y_1, Y_2 :

$$F_{Y_1, Y_2}(y_1, y_2) = P(Y_1 \leq y_1, Y_2 \leq y_2) = P(\{(x_1, x_2) | g_1(x_1, x_2) \leq y_1, g_2(x_1, x_2) \leq y_2\}) = \iint_{\{(x_1, x_2) : g_1(x_1, x_2) \leq y_1, g_2(x_1, x_2) \leq y_2\}} f_{X_1, X_2}(x_1, x_2) dx_1 dx_2$$

Let $u_1 = g_1(x_1, x_2), u_2 = g_2(x_1, x_2)$ InvFT change of vars since $J \neq 0$

$$\therefore F_{Y_1, Y_2}(y_1, y_2) = \iint_{\{(u_1, u_2) : u_1 \leq y_1, u_2 \leq y_2\}} f_{X_1, X_2}(x_1(u_1, u_2), x_2(u_1, u_2)) |J(x_1, x_2)|^{-1} du_1 du_2$$

Clearly, the PDF is the partial derivative w.r.t. y_1, y_2

$$\therefore f_{Y_1, Y_2}(y_1, y_2) = \frac{\partial^2 F_{Y_1, Y_2}}{\partial y_1 \partial y_2} = f_{X_1, X_2}(x_1, x_2) |J(x_1, x_2)|^{-1}, \text{ where } y_1 = g_1(x_1, x_2), y_2 = g_2(x_1, x_2) \quad \square$$

EXAMPLE

Let $Y_1 = X_1 + X_2, Y_2 = X_1 - X_2$

Then, $X_1 = \frac{1}{2}(Y_1 + Y_2), X_2 = \frac{1}{2}(Y_1 - Y_2) \Rightarrow \forall (y_1, y_2) \in \mathbb{R}^2, \exists! (x_1, x_2) \in \mathbb{R}^2$, s.t. $y_1 = g_1(x_1, x_2), y_2 = g_2(x_1, x_2)$, where $g_1(x_1, x_2) = x_1 + x_2, g_2(x_1, x_2) = x_1 - x_2$

$$\therefore J(x_1, x_2) = \frac{\partial(g_1, g_2)}{\partial(x_1, x_2)} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = -2 \neq 0$$

$$\therefore f_{Y_1, Y_2}(y_1, y_2) = f_{X_1, X_2}\left(\frac{1}{2}(y_1 + y_2), \frac{1}{2}(y_1 - y_2)\right) \cdot \frac{1}{|-2|} = \frac{1}{2} f_{X_1, X_2}\left(\frac{y_1 + y_2}{2}, \frac{y_1 - y_2}{2}\right)$$

EXAMPLE

Let $X_1, X_2 \sim N(0, 1)$ be indep. Let $Y_1 = X_1 + X_2, Y_2 = X_1 - X_2$

What can we say about the joint distribution of Y_1, Y_2 ?

Solution

Recall: If $X_1 \sim N(\mu_1, \sigma_1^2), X_2 \sim N(\mu_2, \sigma_2^2)$ indep, then $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

$$\therefore Y_1 \sim N(0, 2), Y_2 \sim N(0, 2)$$

Moreover, by indep, $f_{X_1, X_2}(x_1, x_2) = f_{X_1}(x_1) f_{X_2}(x_2) = \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{x_1^2}{2}}\right) \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{x_2^2}{2}}\right) = \frac{1}{2\pi} e^{-\frac{x_1^2 + x_2^2}{2}}$

$$\begin{aligned} \therefore f_{Y_1, Y_2}(y_1, y_2) &= \frac{1}{2} f_{X_1, X_2}\left(\frac{y_1 + y_2}{2}, \frac{y_1 - y_2}{2}\right) \\ &= \frac{1}{4\pi} e^{-\frac{1}{2} \left[\left(\frac{y_1 + y_2}{2}\right)^2 + \left(\frac{y_1 - y_2}{2}\right)^2 \right]} = \frac{1}{4\pi} e^{-\frac{1}{2} \left[\frac{y_1^2}{2} + \frac{y_2^2}{2} \right]} = \left(\frac{1}{\sqrt{2\pi(2)}} e^{-\frac{y_1^2}{2(2)}}\right) \left(\frac{1}{\sqrt{2\pi(2)}} e^{-\frac{y_2^2}{2(2)}}\right) \\ &= f_{Y_1}(y_1) f_{Y_2}(y_2) \end{aligned}$$

$$\therefore Y_1, Y_2 \text{ are indep } \sim N(0, 2)$$

FACT

If X_1, X_2 : indep identically distributed rand vars and $Y_1 = X_1 + X_2, Y_2 = X_1 - X_2$ are indep, then X_1, X_2 are normally distributed

More generally, given $X_1, X_2, \dots, X_n$, if we def $Y_i = g_i(X_1, \dots, X_n) \forall 1 \leq i \leq n$, s.t. $(X_1, \dots, X_n) \mapsto (Y_1, \dots, Y_n)$ is a bij and $J(x_1, \dots, x_n) \neq 0$, then $f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) = f_{X_1, \dots, X_n}(x_1, \dots, x_n) |J(x_1, \dots, x_n)|^{-1}$

EXPECTATION OF FUNCTION OF RANDOM VARIABLES**RECALL**

If X is a discrete rand var w/ values in $\mathbb{N}_0$, then $\mathbb{E}[X] = \sum_{n=0}^{\infty} n P(X=n)$

LEMMA

Let X be a conti. rand var, then $\mathbb{E}[X] = \int_0^{\infty} P(X \geq t) dt = \int_0^{\infty} P(X \leq -t) dt$

Proof

$$\text{Notice, } \int_0^{\infty} P(X \geq t) dt = \int_0^{\infty} \int_t^{\infty} f_X(x) dx dt = \int_0^{\infty} \int_0^x f_X(x) dt dx = \int_0^{\infty} f_X(x) \int_0^x 1 dt dx = \int_0^{\infty} x f_X(x) dx$$

$$\int_0^{\infty} P(X \leq -t) dt = \int_0^{\infty} \int_{-\infty}^{-t} f_X(x) dx dt = \int_{-\infty}^0 \int_0^{-x} f_X(x) dt dx = - \int_{-\infty}^0 x f_X(x) dx$$

$$\therefore \int_0^{\infty} P(X \geq t) dt - \int_0^{\infty} P(X \leq -t) dt = \int_0^{\infty} x f_X(x) dx - (- \int_{-\infty}^0 x f_X(x) dx) = \int_{-\infty}^{\infty} x f_X(x) dx = \mathbb{E}[X] \quad \square$$

PROPOSITION

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Let X_1, X_2 be rand vars on a prob space and $g: \mathbb{R}^2 \rightarrow \mathbb{R}$

Then, (discrete) $\mathbb{E}[g(X_1, X_2)] = \sum_{x_1, x_2} g(x_1, x_2) P_{X_1, X_2}(x_1, x_2)$

(continuous) $\mathbb{E}[g(X_1, X_2)] = \iint_{\mathbb{R}^2} g(x_1, x_2) f_{X_1, X_2}(x_1, x_2) dx_1 dx_2$

Proof of Cont. Case

By lemma, $\mathbb{E}[g(X_1, X_2)] = \int_0^\infty P(g(X_1, X_2) \geq t) dt - \int_0^\infty P(g(X_1, X_2) \leq -t) dt$

Notice, $P(g(X_1, X_2) \geq t) = \iint_{\{(x_1, x_2): g(x_1, x_2) \geq t\}} f_{X_1, X_2}(x_1, x_2) dx_1 dx_2$

$$\begin{aligned} \int_0^\infty P(g(X_1, X_2) \geq t) dt &= \int_0^\infty \iint_{\{(x_1, x_2): g(x_1, x_2) \geq t\}} f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 dt = \iint_{\{(x_1, x_2): g(x_1, x_2) \geq 0\}} \int_0^{g(x_1, x_2)} f_{X_1, X_2}(x_1, x_2) dt dx_1 dx_2 \\ &= \iint_{\{(x_1, x_2): g(x_1, x_2) \geq 0\}} g(x_1, x_2) f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 \\ \int_0^\infty P(g(X_1, X_2) \leq -t) dt &= \int_0^\infty \iint_{\{(x_1, x_2): g(x_1, x_2) \leq -t\}} f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 dt = \iint_{\{(x_1, x_2): g(x_1, x_2) \leq 0\}} \int_0^{-g(x_1, x_2)} f_{X_1, X_2}(x_1, x_2) dt dx_1 dx_2 \\ &= - \iint_{\{(x_1, x_2): g(x_1, x_2) \leq 0\}} g(x_1, x_2) f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 \\ \therefore \mathbb{E}[g(X_1, X_2)] &= \iint_{\{(x_1, x_2): g(x_1, x_2) \geq 0\}} g(x_1, x_2) f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 - (- \iint_{\{(x_1, x_2): g(x_1, x_2) \leq 0\}} g(x_1, x_2) f_{X_1, X_2}(x_1, x_2) dx_1 dx_2) \\ &= \iint_{\mathbb{R}^2} g(x_1, x_2) f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 \quad \square \end{aligned}$$

EXAMPLE

Consider a road of length L .

An accident occurs at X , unif along the road.

An ambulance is at Y , unif along the road, indep of X .

How far must the ambulance travel to get to the accident? What is the expected distance?

Solution

$X \sim \text{Unif}([0, L])$, $Y \sim \text{Unif}([0, L])$, indep

$$\therefore f_{X, Y}(x, y) = \begin{cases} \frac{1}{L^2}, & 0 \leq x, y \leq L \\ 0, & \text{o/w} \end{cases}$$

Notice, $\text{dist}(X, Y) = |X - Y|$

$$\begin{aligned} \text{Thus, } \mathbb{E}[\text{dist}(X, Y)] &= \iint_{\mathbb{R}^2} |X - Y| f_{X, Y}(x, y) dx dy = \frac{1}{L^2} \int_0^L \int_0^L |x - y| dx dy = \frac{1}{L^2} \int_0^L \left(\int_0^y y - x dx + \int_y^L x - y dx \right) dy \\ &= \frac{1}{L^2} \left(\frac{1}{2} y^2 + \frac{1}{2} L^2 - Ly + \frac{1}{2} y^2 \right) dy = \frac{1}{L^2} \int_0^L (y^2 + \frac{1}{2} L^2 - Ly) dy \\ &= \frac{1}{L^2} \left(\frac{1}{3} y^3 + \frac{1}{2} L^2 y - \frac{1}{2} Ly^2 \right) \Big|_0^L = \frac{1}{L^2} \left(\frac{1}{3} L^3 \right) = \frac{L}{3} \end{aligned}$$

APPLICATION (LINEARITY)

$$\text{If } g(X, Y) = aX + bY, \text{ then } \mathbb{E}[aX + bY] = \iint_{\mathbb{R}^2} g(x, y) f_{X, Y}(x, y) dx dy = \iint_{\mathbb{R}^2} (ax + by) f_{X, Y}(x, y) dx dy = a \int_{\mathbb{R}} x \int_{\mathbb{R}} f_{X, Y}(x, y) dy dx + b \int_{\mathbb{R}} y \int_{\mathbb{R}} f_{X, Y}(x, y) dx dy = a \int_{\mathbb{R}} x f_X(x) dx + b \int_{\mathbb{R}} y f_Y(y) dy \\ = a \mathbb{E}[X] + b \mathbb{E}[Y]$$

APPLICATION (MONOTONICITY)

Let X, Y : rand vars on a prob space. If $X \geq Y$ always then $\mathbb{E}[X] \geq \mathbb{E}[Y]$

Proof by lin of exp. $\mathbb{E}[X] - \mathbb{E}[Y]$

$$X \geq Y \Rightarrow X - Y \geq 0 \Rightarrow \mathbb{E}[X - Y] \geq 0 \Rightarrow \mathbb{E}[X] \geq \mathbb{E}[Y] \quad \square$$

APPLICATION (UNION BOUND/BOOLE'S INEQUALITY)

If $A_1, \dots, A_n$ are events, then $P(\bigcup_{i=1}^n A_i) \leq \sum_{i=1}^n P(A_i)$

Proof

Let X_i be the indicator for A_i , i.e. $X_i = \begin{cases} 1, & A_i \text{ occurs} \\ 0, & A_i \text{ does not occur} \end{cases}$

Then, $\mathbb{E}[X_i] = 1P(A_i) + 0P(A_i^c) = P(A_i)$ by lin of exp.

Let $X = \sum_{i=1}^n X_i$ = # events that occur

Let Y be the indicator for $\bigcup_{i=1}^n A_i$, i.e. $Y = \begin{cases} 1, & x \geq 1 \\ 0, & x = 0 \end{cases}$

Obviously, $Y \leq X$ always

$$\text{By monotonicity, } P(\bigcup_{i=1}^n A_i) = \mathbb{E}[Y] \leq \mathbb{E}[X] \stackrel{\text{lin of exp}}{=} \sum_{i=1}^n \mathbb{E}[X_i] = \sum_{i=1}^n P(A_i) \quad \square$$